EE3 Detailed design

Embodiment design W2-9

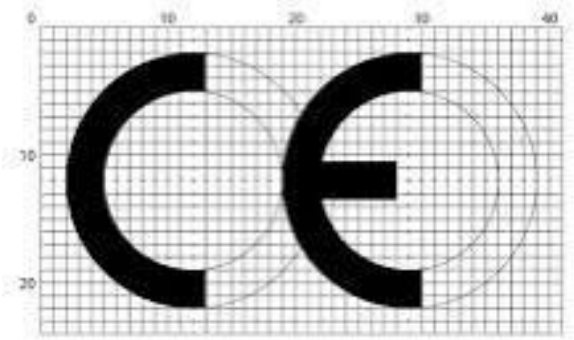
Digital twin W1-12

Detailed design
S1 W11 – S2 Easter

Team defense - After Easter

Prof. skills – After Easter

Technical file



- What is a technical file?
- https://www.cemarkingauthority.com/ce-directives/technical-construction-files
- https://www.conformance.co.uk/ce-marking-assessment-specialists/76-technical-file-information
- Technical file for EE3 will include (https://www.cemarkingauthority.com/ce-directives/technical-construction-files)

  - *Risk assessments (ISO 12100, EN 14121, Risk reduction according to priority list) ER*
  - *Technical description. ER*
  - *Bill of materials (FLoC). ER*
  - Drawings, circuit diagrams, photos, etc TF
  - Specification and, where applicable, Declarations of Conformity for the critical components and materials used. TF
  - Details of any design calculations. TF, some might be necessary for the ER)
  - ~~Test reports and/or assessments.~~
  - Original Manual (very limited for EE3) ER/TF
  - Copy of the Declaration of Conformity. TF
- The user instructions can be greatly reduced for EE3: only the very essential instructions will be added, e.g. building sequence of the machine, as that is also needed to guarantee a proper design.
- A usable (extended) template can be found here: https://iosh.com/media/4539/iosh-theatre-advisory-technical-file-checklist-2019.pdf

Outcome detailed design

- Excel file
  - Detailed description of all parts (material, size, bought or made, …)
  - Cost calculation of all parts (material costs, production costs, …)
  - Excel template is available
- Technical file
  - Mechanical calculations and motor selection + manufacturing plan
  - Technical drawings
  - Electric diagrams/engineering
  - Digital twin Movie and files

➢ **Each student will claim part of the work as "main responsible"** in a separate document, template is available. That does not mean that the rest of the team is no longer (also) responsible for that part.

How to report on mechanical calculations

- https://libguides.newcastle.edu.au/writing-paragraphs/types
- https://libguides.newcastle.edu.au/ld.php?content_id=39162655

Structure of your document:

- Introduction on the general problem (PaStr: Describe)
- General description of your solution (PaStr: Describe)
- Explanation on which problems you will research and why (PaStr: Explain)
- For each problem (= each graph you make)
  - Definition of the problem you are researching (PaStr: Define)
  - The graph you created
  - Describe what is to be learned from the graph (PaStr: Describe)
  - Argue why you trust the result of your graph (PaStr: Argue)
  - Illustrate what the impact is on the design (PaStr: illustrate)
- Final conclusion (PaStr: Argue)

EVA EE3 Mechatronics 2025 Mechanical calculations

- Mechanical calculations and motor selection
  - Realistic design/selection of mechanical components and actuators
  - Manufacturing plan of engineered components
  - Cost estimation
- End Result
  1. (*pass/fail*) Complete set of all relevant technical drawings (NO standard components)
  2. (*pass/fail*) Clear assembly drawing(s) showing coherence and operation of design
  3. Calculations and selection/design of critical components (prove that the design works)
  4. Consistency in mechanical design (correct tolerances, dimensions, materials, …)
  5. Summary of the total cost estimate and breakdown (potential for savings?)
  6. Oral defense

Evaluation of the mechanical design

Below you find the evaluation criteria for Mechanical design in EE2, so a minimal requirement for EE3: When grading, teaching staff will start from the sketches and technical drawings available and only then switch to the annotations made to these drawings. A “Drawings and sketches first, text second" approach will be used. From a design perspective the following items will be graded:

- Degrees of freedom. Is every component correctly constrained/unconstrained in all degrees of freedom? Is movement (rotation and translation) permitted or avoided in the required direction(s)? (Tip: it’s not too different from a CAD assembly)
- Friction vs geometric design. Is the choice between friction and geometric based design well made?
- Proper selection on the number of functions a component fulfills. Is it clear why functions are (not) combined into one (multiple) components? (Tip: free body diagrams are very useful to explain choices made)
- Robustness of the design. Is the design expected to behave robustly? Is it evident how all forces and torques involved are dealt with and why?
- Positional correctness of the design. Are all components positioned with proper accuracy? Does the design allow for the adaptations necessary during testing? (Remember that we are building a prototype not a series-product. With regards to “Proper”, this does not equal “very precise”.)
- Ease of production and proper use of DIN standards. Are different components designed with available production methods in mind? How are e.g. production errors from laser cutting taken into account? How does the design itself allow/compensate for production errors?
- Modularity. Is the design made using standardization / recurring components in such a way that different students can work on different parts of the design in parallel?
- Creativity/innovation/references? Does the design hold creative and or innovative solutions? Is the design appropriately based on existing devices and on-line sources? Has sufficient attention been spend on the intricate details of existing designs?

EVA EE3 Digital Twin – PLC

Assignment 4:

- Digital Twin video demonstrating the MCD model is working
  - Preliminary design, changes can be made afterwards (improved design, CAD, …)
  - All joints, motors,… to prove the concept
  - Exact specs, TIA,… don't need to be implemented yet!

EVA EE3 Digital Twin – PLC, final result

- Create a Digital Twin of your machine
  1. Define a research question
  2. Include all necessary mechanics (joints, motors, … , based on specs)
  3. Working control program (fully automated program)
  4. Conclusions Digital Twin
     - Test your model
     - Prove it is working as expected or conclude what needs to be changed/improved ==> Iterations not necessary
- End result
  1. Video working model + explanation
  2. TIA + NX files
  3. Report Digital Twin (setup, conclusions)

Automation in EE3: use 12100 & 13849 but skip pl achieved.

- An engineer makes sure a machine is designed according to the Belgian law, in particular the Machinery Directive.
- Risk analysis and reduction according to priorities. Use of multiple safety functions is recommended, e.g. SLS when approaching, SS1 when entering danger zone. Design the electric diagrams according to the results of the RA/RR.
- The system is engineered first, and then drawn in SEE, or a similar software. We suggest to use SEE electrical.
  A drawing template is available, but its not mandatory to use it. The diagrams are engineered according to EN 60204, and the "Group T company standard" as used in the examples in OIS/DICS.
- Based on the risk analysis (see DICS/OIS, EN 12100), a dual channel or single channel E-stop relay is implemented according to EN 13849. In the text this choice is justified. All "dangerous" actuators are interfaced correctly (to the SRP/CS).
  Proper safety functions (e.g. SS1, SLS, …) and detections (e.g. light curtain) are selected and implemented in the design.
  The safety system will still allow economically feasible and practical operation of the machine.
- All components in the diagrams have a unique name, according to EN 81346. In the report that same name is used consistently to refer to that component. The terminals on the component are named on the drawing as on the actual component, even if the manufacturer did not respect a relevant standard.
- The electrical system should be wired to terminal strips, so that each cable can easily be connected to a terminal (strip) on site. A bus bar and/or terminal strips are used to distribute potentials (e.g. the 24 VSV). Label the cables on the diagrams as well.
- Remote I/O cabinets, or remote I/O systems are used if necessary. Field bus interfaces are preferred over analog/discrete signals. The control system used should be suitable for the design (e.g. not an Arduino for an industrial machine).
- The functions described in the text should match the system represented in the circuit diagrams (e.g. an asynchronous motor needs a sensor, or it will not stop when a certain position is reached; A position detection switch cannot report an analog value.)

Evaluation detailed design P.Geenen

11 & 12 are main EVA criteria, use of **EN 12100 & 13849 is mandatory**, **plan** but no calculation of PL achieved.

1.	Does the control system works **conceptually**?	**Own work**?	**RDS?**	Pass / Fail
2.	RA encompasses **all life phases**? Does **it cover all risks**, clear but not too detailed?			① ② ③ ④ ⑤
3.	Major risks discovered through methodical **risk assessment** via **risk graph**?			① ② ③ ④ ⑤
4.	Is it clear from the text that the **risk reduction was done according to priority** ?			① ② ③ ④ ⑤
5.	Justification of **residual risk** and **reductions considered** but not applied.			① ② ③ ④ ⑤
6.	**Indicated where RR depends on SRP/CS? Multiple safety functions(SF)? PL_r for each SF**?			① ② ③ ④ ⑤
7.	Made a **plan to achieve each PL_r? Proper components selected? Implemented correctly?**			① ② ③ ④ ⑤
8.	SEE: Components clearly represented; **order codes where necessary** (standard cables!)?			① ② ③ ④ ⑤
9.	SEE: All supply / interfacing correct? Logical use of terminals and cables?			① ② ③ ④ ⑤
10.	TF complete? Added most important datasheets(pages)? PL/SIL indicated?			① ② ③ ④ ⑤
11.	**Justification for the choices and components (oral).** *B/M ± 4*			① ② ③ ④ ⑤
12.	**Per actuator: Correct connections? Proper SRP/CS for every dangerous actuator? Suitable sensors & drives for each actuator for the proposed functionality?** */2 each*			
	○ SEE: Actuator 0 (-M	):sensors, cables, drive, E-stop?		① ② ③ ④ ⑤
	○ SEE: Actuator 1 (-M	):sensors, cables, drive, E-stop?		① ② ③ ④ ⑤
	○ SEE: Actuator 2 (-M	):sensors, cables, drive, E-stop?		① ② ③ ④ ⑤
	○ SEE: Actuator 3 (-M	):sensors, cables, drive, E-stop?		① ② ③ ④ ⑤

Embodiment

Safety functions - Definitions	SF1	SF2	SF3	SF4	SF5	SF6	SF7
Brief description	Emergency stop	Electronically interlocking function	Perimeter monitoring	Safe brake control (SBC)	Safe brake test (SBT)	Enabling function	Safe limited speed (SLS)
Triggering Event	Push of any emergency stop button	Object detected in Hazard zone 1 under the machine, in direct reach of the machine	Object entering in hazard zone 2 around the machine	Emergency stop or any other stop	Violation of the SBT procedure	release both enabling test buttons	Violation of the limited speed
Safety-related reaction	Controlled stop (SS1)	Controlled stop (SS1)	Controlled stop (SS1)	Uncontrolled stopping (STO)	Uncontrolled stopping (STO)	Controlled stop (SS2)	Controlled stop (SS1)
Dangerous part of the machine	The three moving joints of the machine	The three moving joints of the machine	The three moving joints of the machine	The three moving joints of the machine	The three moving joints of the machine	The three moving joints of the machine	The three moving joints of the machine
Operating mode	Any mode	Automatic mode	Automatic mode	Any mode	Any mode	Setup or maintenace mode	Setup or maintenace mode
PL_r	d	d	b	d	d	c	c
Demand rate	20x per hour	1x per hour	10x per hour	20x per hour	1x per day	1x per hour	1x per day
Reaction time	Controller(46,8ms) motor (500-1000ms)	Controller(154,8ms) motor (500-1000ms)	Controller(126,8ms) motor(500-1000ms)	Motor(500-1000ms)	Motor(500-1000ms)	Motor(500-1000ms)	Motor(500-1000ms)
Behaviour in the event of a power failure	Uncontrolled stopping (STO) normally activated brakes on all motors	Uncontrolled stopping (STO) normally activated brakes on all motors	Uncontrolled stopping (STO) normally activated brakes on all motors	Uncontrolled stopping (STO) normally activated brakes on all motors	Uncontrolled stopping (STO) normally activated brakes on all motors	Uncontrolled stopping (STO) normally activated brakes on all motors	Uncontrolled stopping (STO) normally activated brakes on all motors
Priority	1 (highest priority)	4 (with lower priority than the Safe brake control function)	4 (with lower priority than the Safe brake control function)	3 (With lower priotity than the Safe brake test function)	2(with lower priority than the emergency-stop function)	5 (with lower priority than SLS)	4 (with lower priority than the Safe brake control function)
Supplementary SF	Electronically interlocking function, perimeter monitoring and safe brake control (SBC)	Safe brake control (SBC)	Safe brake control (SBC)	Safe brake test (SBT)	-	Safe Limited Speed (SLS)	-
Remarks	When SS1 is finished mechanical brakes are activated. These need power to be deactivated (NC)	When SS1 is finished mechanical brakes are activated. These need power to be deactivated (NC)	When SS1 is finished mechanical brakes are activated. These need power to be deactivated (NC)	Safe brake control should thake priority over any function that relies on this. The emergency stop is an exception in worst case with this initiate STO	-		-
Remarks on PLr	S2, F2, P1: Avoiding the hazard of not being able to stop, is in a rare case possible by interrupting the perimeter monitoring	S2, F2, P1: The safety scanners simply cannot fale. These are worst case scenario safety devises. Nobody without extensive training can enter Hazard zone 1	S1, F2, P1: Individuals are not in immediate danger when entering Hazard Zone 2; however, the likelihood of accidents increases when access to this area is permitted.	S2, F1, P2: Although the machine will be stopped often, the hazardous moment is only when the machine is stopped, which is small compared to other hazards	S2, F1, P2: Safe brake test should had the same performance level als Safe brake control in order to fully trust both systems	S2, F1, P1: Individuals could get severly hurt if tempered with the enabling function.	S2, F1, P1: Individuals could get severly hurt if tempered SLS.

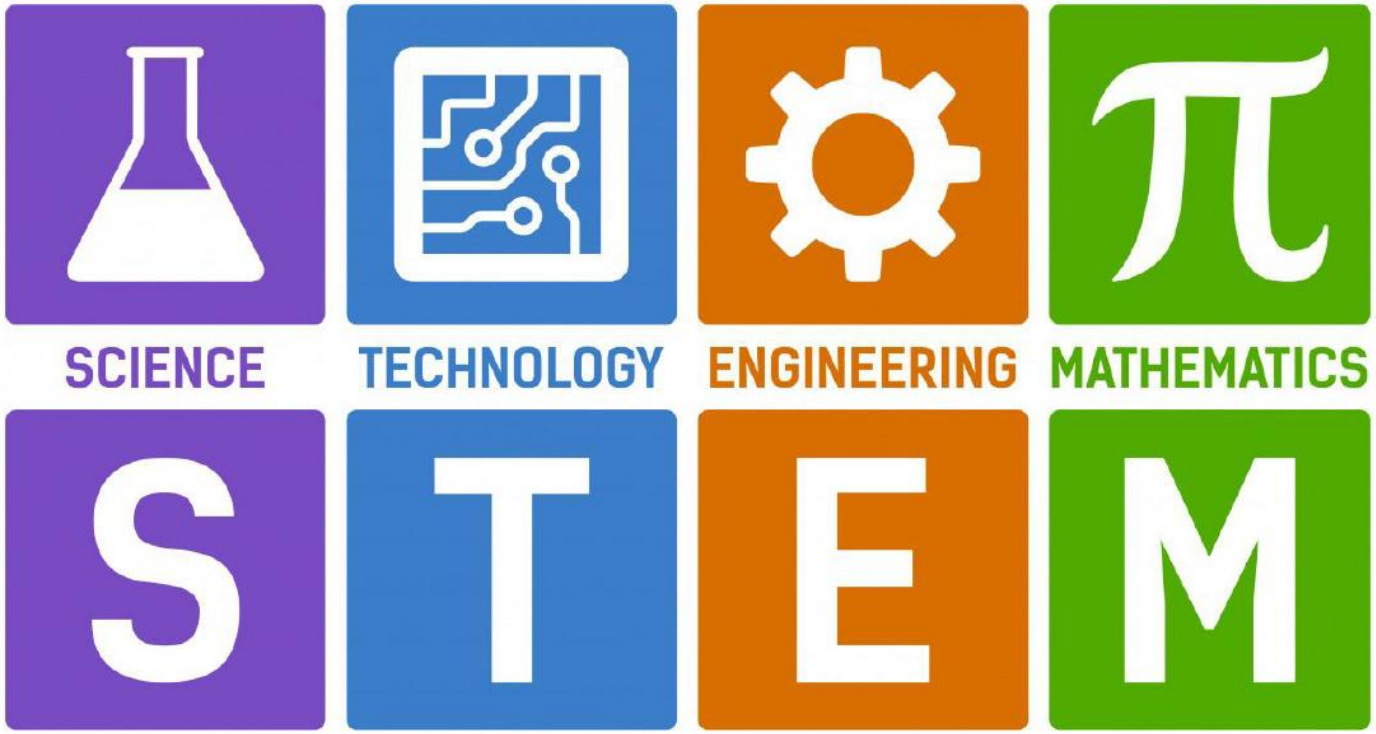
Just an example

Created by Pieter Nagels & Seppe Huybrechts

Project overview:

- Now that your design is finished: can you elaborate on the real limits/performance/specs of your design? The assignment mentioned 3 mm repeatability, and neglectable forces and moments on the end actuator. Now that your design is finished: can you elaborate on the real limits/performance/specs of your design?
- The final design should be complete (see eva slide) and should update the cost of each component in the required format, as well as include other costs. The project and professional competences = 6 ects · 27,5 h/ects · 50 €/h · 4 people = 33 k€ result is what we expect.
  - **The final design is crucial for your score on the project!**
  - **We will closely monitor and evaluate your progress and teamwork during these weeks.**

➢ ***Consistent use of RDS:*** *Pass/fail*
Fail means that you will not pass EE3 this year. Can we be any clearer on that?

Professional skills

- Writing
- Presenting to pre-bachelors
- Open projects
- …